

Cosmic Magnetic Shear of Blazars

Center for Computational Astrophysics – Perimeter Institute Cosmology Group Meeting

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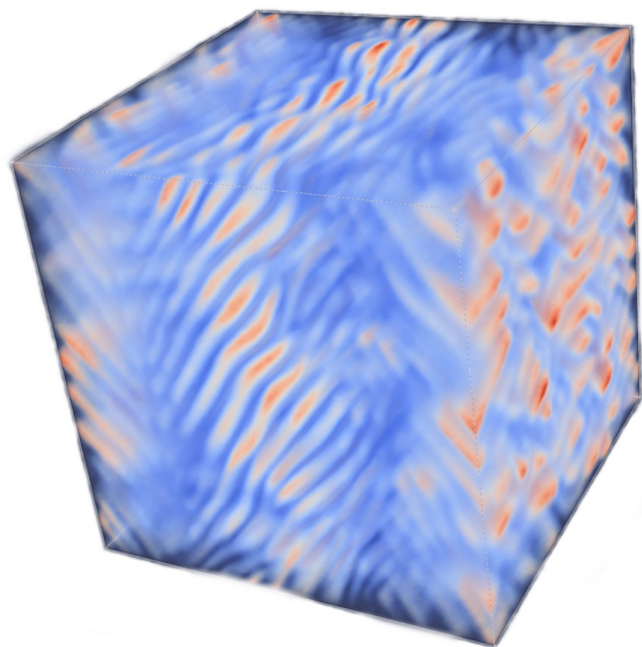


Roadmap

- I will...
- I. **Review** the present status of intergalactic magnetic fields (IGMFs) and blazar halos.
 - II. **Signal**: Show that IGMFs induce correlated image distortions of blazars.
 - III. **Noise**: Quantify and describe systematics and null tests for such distortions.
 - IV. **Comment** on other blazar-based methods of probing IGMFs.

Fermi LAT is sensitive to IGMFs with strengths $\sim(10^{-16}-10^{-14})$ G and coherence lengths $\sim(1-100)$ Mpc for a variety of spectral indices.

IGMFs Broadly: What and Why?



$$(B_0, \lambda_B, n_B)$$

[Zhou, Zhdankin, Kunz, Loureiro, Uzdensky (2022)]

Galactic Seeds



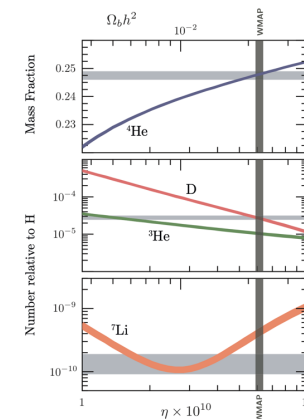
[NASA, SOFIA, HAWC+, Alejandro S. Borlaff;
JPL-Caltech, ESA, Hubble]

Inflation



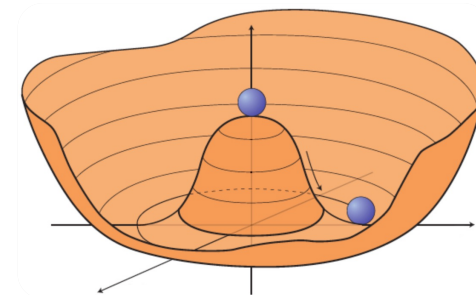
[Faucher-Giguère, Lidz, Hernquist (2008)]

Baryogenesis



[Baumann]

Phase Transitions

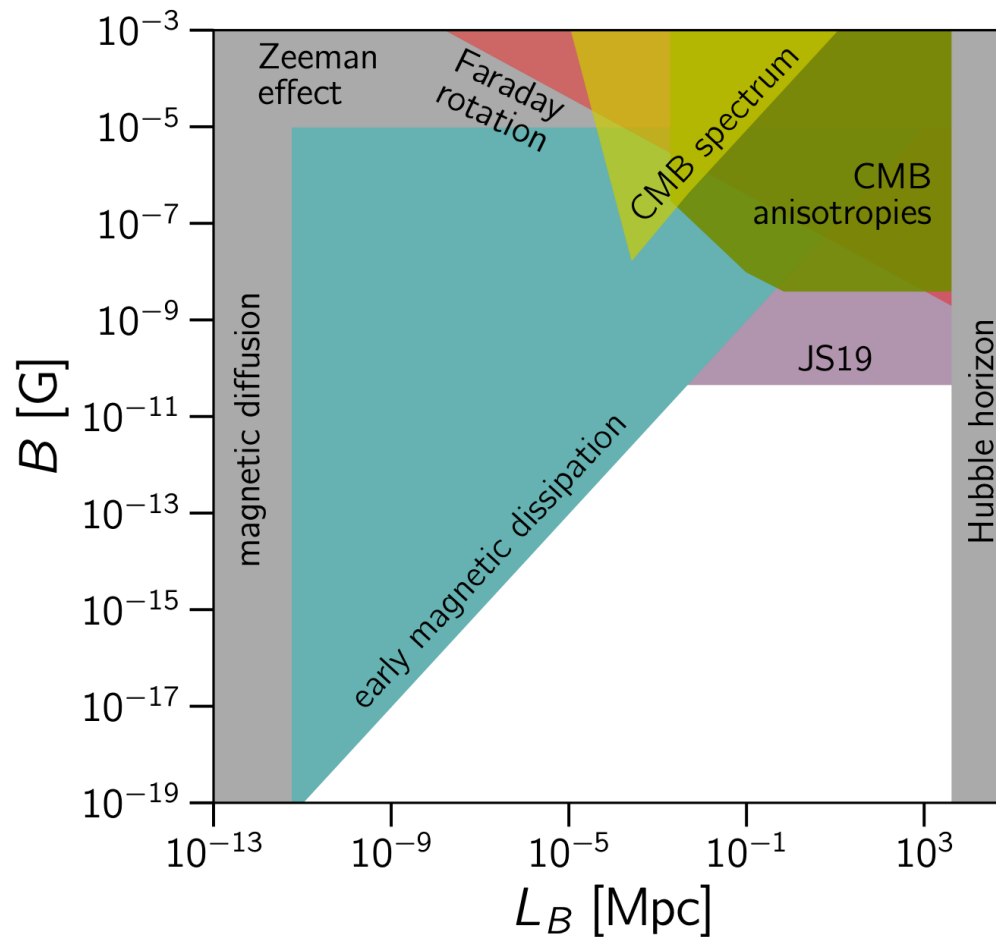


[Alvarez-Gaume, Ellis (2010)] ³

I. Review

IGMFs: A Mini Review of Constraints (Part I)

[Batista & Saveliev (2021)]



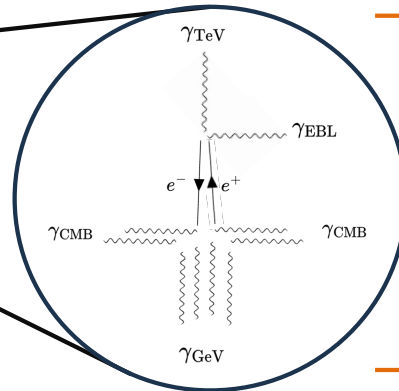
IGMFs & Blazars

IGMF ✗

★ $E_{\gamma}^{\text{bl}} \sim \text{TeV}$

$$\lambda_{\gamma\gamma}(z, E) = \frac{800 \text{ Mpc}}{(1+z)^2} \left[\frac{1 \text{ TeV}}{E_{\gamma}^{\text{bl}}(E)} \right]$$

$$E_e \sim E_{\gamma}^{\text{bl}}/2$$

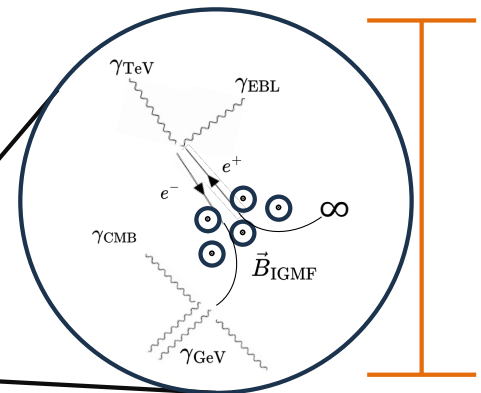


$$t_{\text{IC}}(|\mathbf{x}|, E) = 0.43 \text{ Mpc} \left[\frac{0.5 \text{ TeV}}{E_e(E)} \right] \left[\frac{1}{1 + z_{\gamma\gamma}(|\mathbf{x}|)} \right]^4$$

[Nishikov (1961)]
[Jelley (1966)]
[Gould and Schröder (1967)]

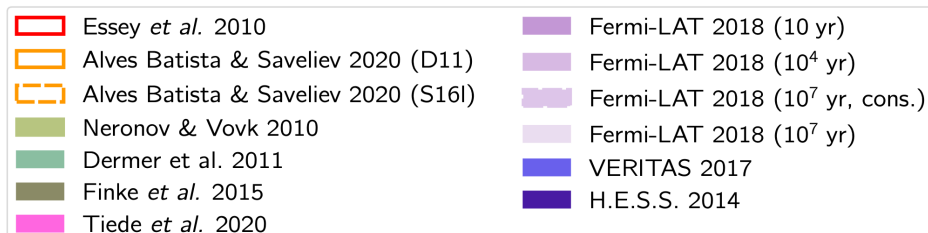
IGMF ✓

★



[Aharonian, Coppi, and Völk (1993)]

IGMFs: A Mini Review of Constraints (Part II)



1) Blazar Spectral Cutoffs

[Biteau *et al.* (2020)]

2) IGMFs

[Fermi LAT (2018)]

[Blanco, Ghosh, Jacobsen, & Linden (2023)]

3) BSM Physics

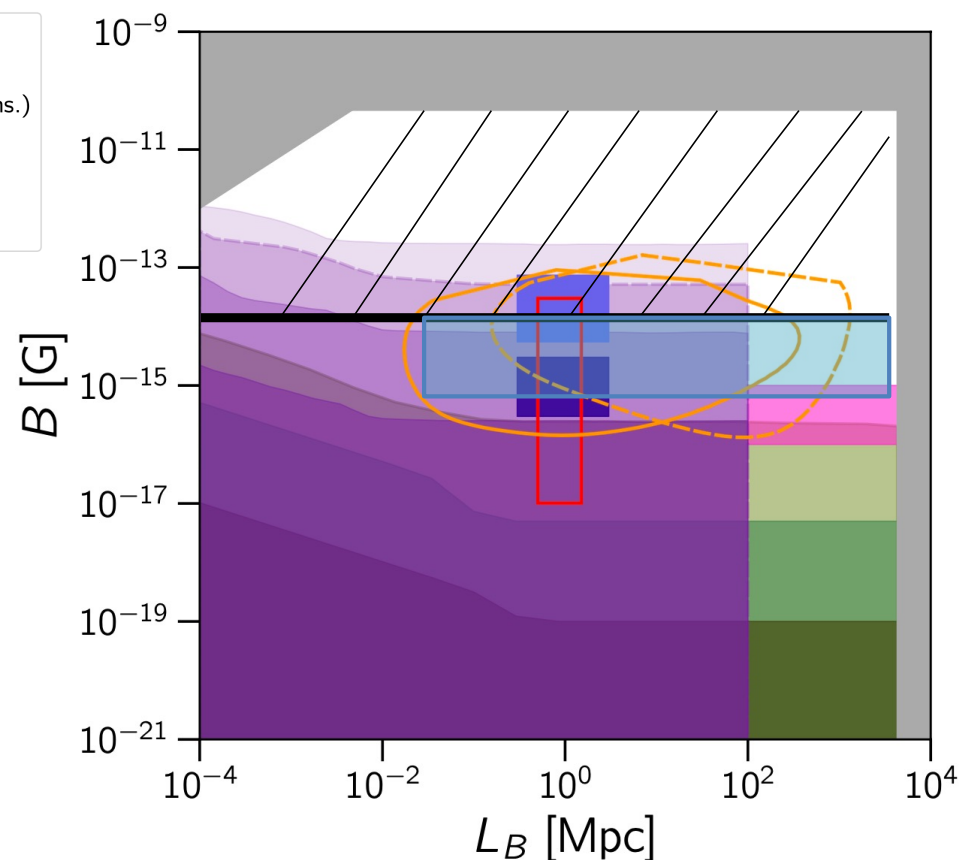
[Dessert, Dunskey, & Safdi (2022)]

[Caputo, Liu, Mishra-Sharma, & Ruderman (2020)]

4) Plasma Instabilities

[Broderick, Chang, & Pfrommer (2011)]

[Sironi & Giannios (2013)]

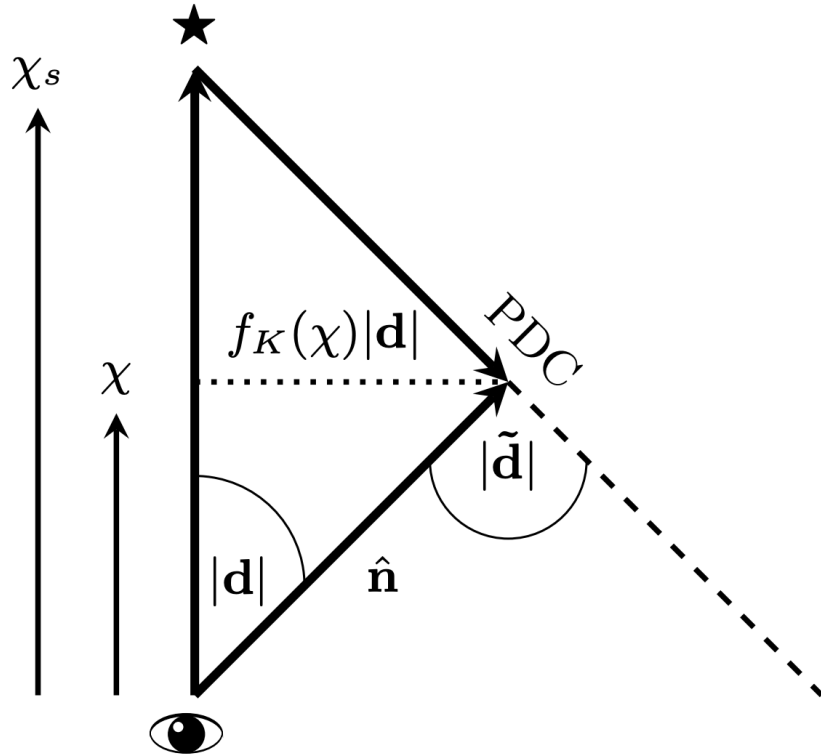


[Batista & Saveliev (2021)]

[Blanco, Ghosh, Jacobsen, & Linden (2023)]

II. Signal

Magnetic Lensing of Blazars



$$\tilde{\mathbf{d}}(\hat{\mathbf{n}}, E) = \frac{\delta \mathbf{p}_e}{|\mathbf{p}_e|} = -\frac{\hat{\mathbf{n}} \times \mathbf{B}(\mathbf{x}, t)}{B_c(|\mathbf{x}|, E)}$$

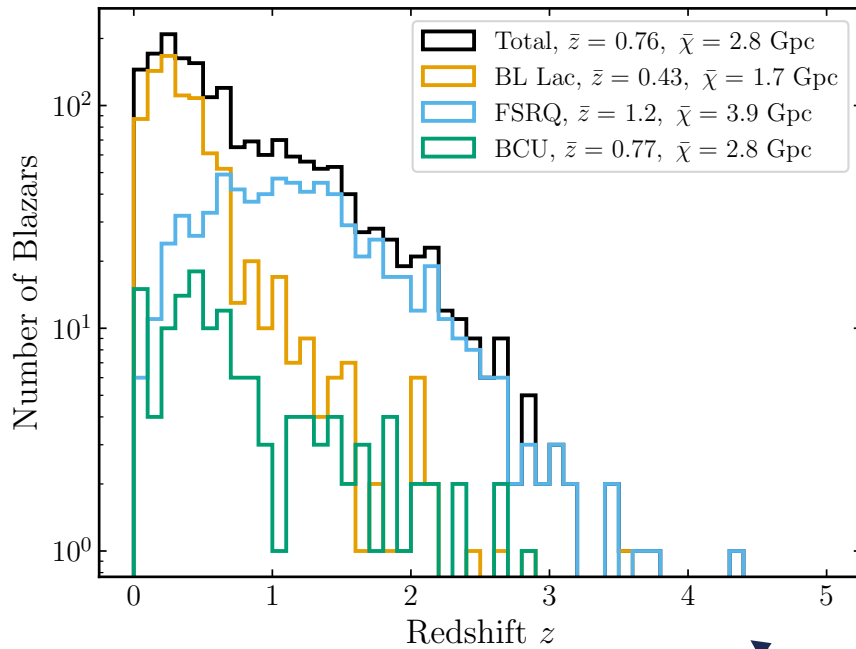
$$B_c = 1.5 \times 10^{-15} \text{ G} \left[\frac{E_e(E)}{0.5 \text{ TeV}} \right] \left(\frac{1 e}{|q|} \right) \left[\frac{0.43 \text{ Mpc}}{t_{\text{IC}}(|\mathbf{x}|, E)} \right]$$

$$\mathbf{d}(\hat{\mathbf{n}}, E) = - \int_0^{\chi_{\text{lim}}} d\chi P(\chi, E) \frac{[\hat{\mathbf{n}} \times \mathbf{B}(\chi, \hat{\mathbf{n}})]}{B_c(\chi, E)}$$

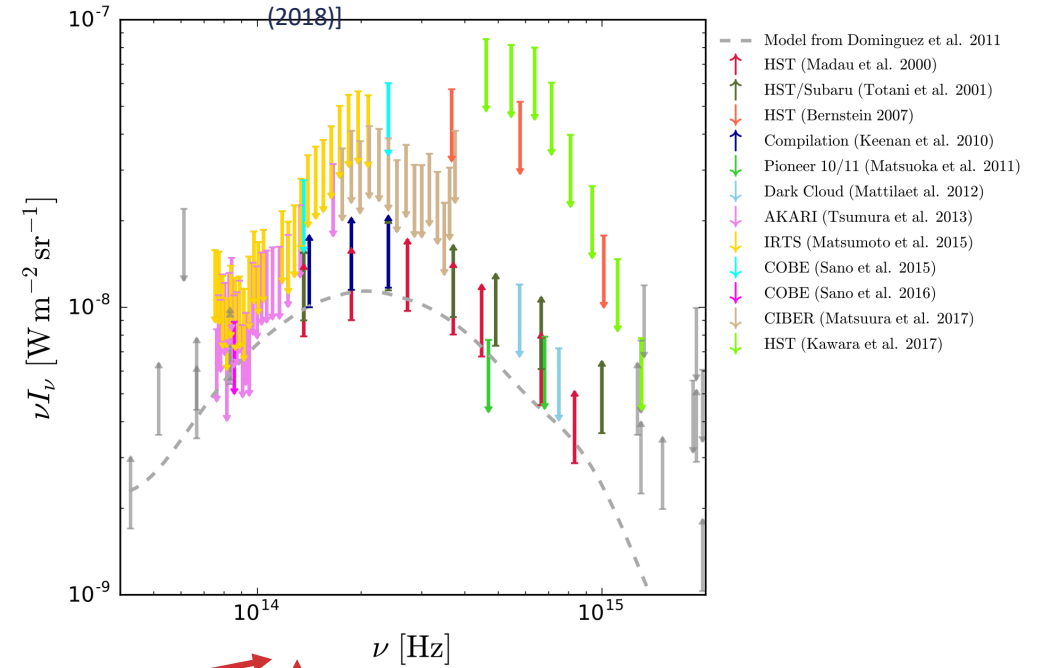
$$P(\chi, E) = \frac{d\tau_{\gamma\gamma}}{d\chi} \int_{\chi}^{\chi_{\text{lim}}} d\chi_s \frac{dn}{d\chi_s} \frac{f_K(\chi_s - \chi)}{f_K(\chi_s)} e^{-\tau_{\gamma\gamma}(\chi, \chi_s, E)}$$

Astrophysical Inputs

[Ajello et al., Fermi Collaboration DR3 (2022)]



[Hill, Masui, & Scott
(2018)]



$$P(\chi, E) = \frac{d\tau_{\gamma\gamma}}{d\chi} \int_{\chi}^{\chi_{\text{lim}}} d\chi_s \frac{dn}{d\chi_s} \frac{f_K(\chi_s - \chi)}{f_K(\chi_s)} e^{-\tau_{\gamma\gamma}(\chi, \chi_s, E)}$$

Cosmic Shear Review

A. Weak Lensing Parameterization

$$A_{ij} = g_{ij} + \nabla_i d_j = (1 - \kappa) g_{ij} + \omega \epsilon_{ij} - \gamma_{ij}$$

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}, \quad \epsilon_{ij} = \begin{pmatrix} 0 & \sin \theta \\ -\sin \theta & 0 \end{pmatrix},$$

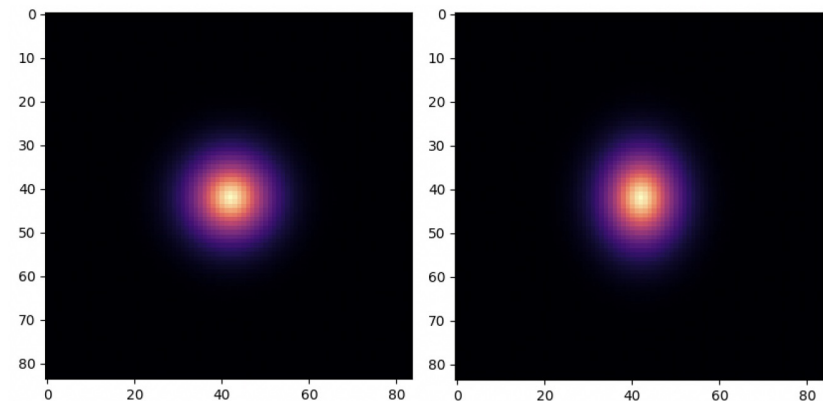
$$\gamma_{ij} = \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{pmatrix}.$$

B. Measurement

$$Q_{ij} = \frac{\int d^2\theta I(\theta) \Delta\theta_i \Delta\theta_j}{\int d^2\theta I(\theta)}$$

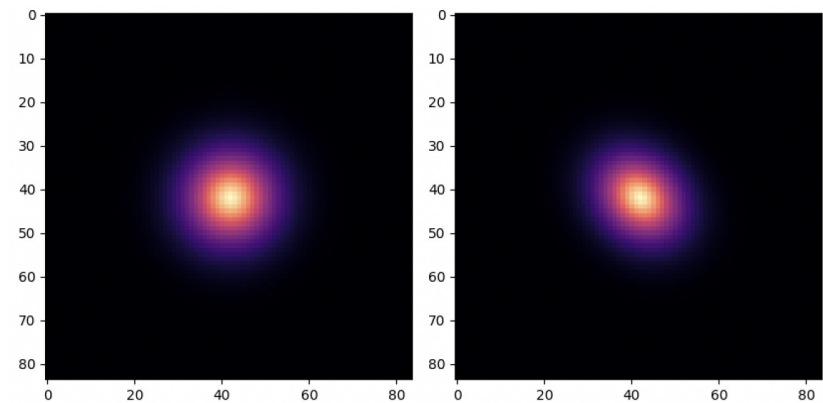
$$\gamma_1 = \frac{1}{2} \frac{Q_{11} - Q_{22}}{Q_{11} + Q_{22}}, \quad \gamma_2 = \frac{1}{2} \frac{Q_{12} + Q_{21}}{Q_{11} + Q_{22}}$$

[Stebbins (1996)] [Kilbinger (2014)]



(a) Unlensed

(b) γ_1



(e) κ

(f) γ_2

[Philcox, König, Alexander, & Spergel (2024)]

E- & B- Mode Decomposition

Deflection Angles

$$\mathbf{d}(\hat{\mathbf{n}}) = \nabla\varphi(\hat{\mathbf{n}}) + \hat{\mathbf{n}} \times \nabla\psi(\hat{\mathbf{n}}) = \hat{\mathbf{n}} \times \boldsymbol{\delta}(\hat{\mathbf{n}})$$

$$\boldsymbol{\delta}(\hat{\mathbf{n}}) = [\nabla \times \boldsymbol{\Phi}(\hat{\mathbf{n}}) + \nabla\psi(\hat{\mathbf{n}})]$$

$$\boldsymbol{\Phi}(\hat{\mathbf{n}}) = \varphi(\hat{\mathbf{n}}) \hat{\mathbf{n}}$$

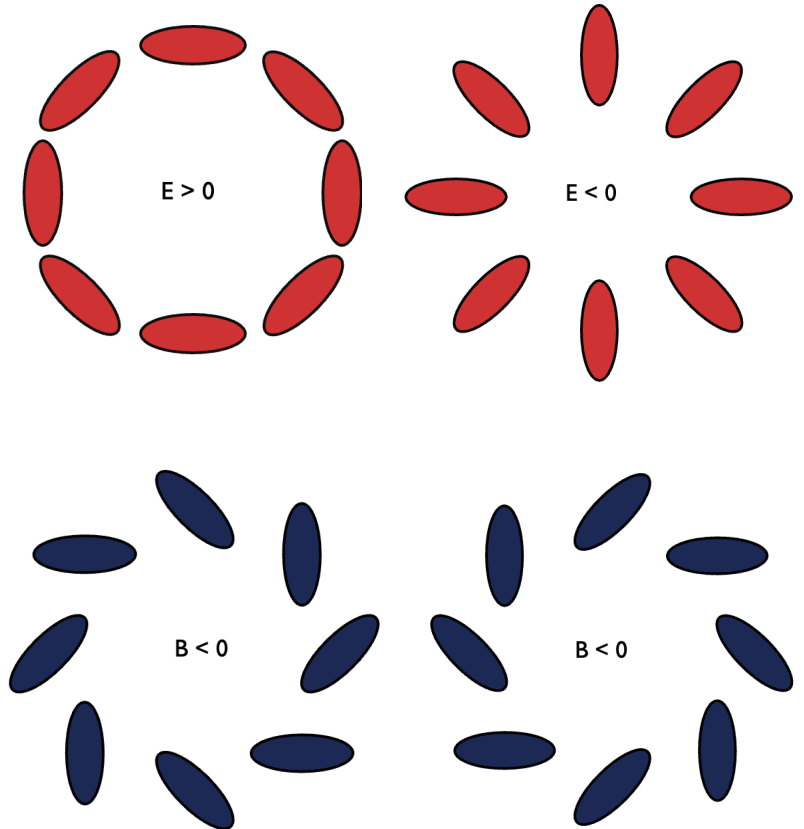
Lensing Potentials

$$\nabla^2\varphi(\hat{\mathbf{n}}) = \nabla \cdot \mathbf{d}(\hat{\mathbf{n}}) = \hat{\mathbf{n}} \cdot [\nabla \times \boldsymbol{\delta}(\hat{\mathbf{n}})]$$

$$\nabla^2\psi(\hat{\mathbf{n}}) = \nabla \cdot \boldsymbol{\delta}(\hat{\mathbf{n}}) = \hat{\mathbf{n}} \cdot [\nabla \times \mathbf{d}(\hat{\mathbf{n}})]$$

Decomposition

$$\gamma_{(\ell m)}^{\text{E}} = N_{\ell}\varphi_{(\ell m)}, \quad \gamma_{(\ell m)}^{\text{B}} = N_{\ell}\psi_{(\ell m)}$$



[Yamauchi, Namikawa, & Taruya (2012)]

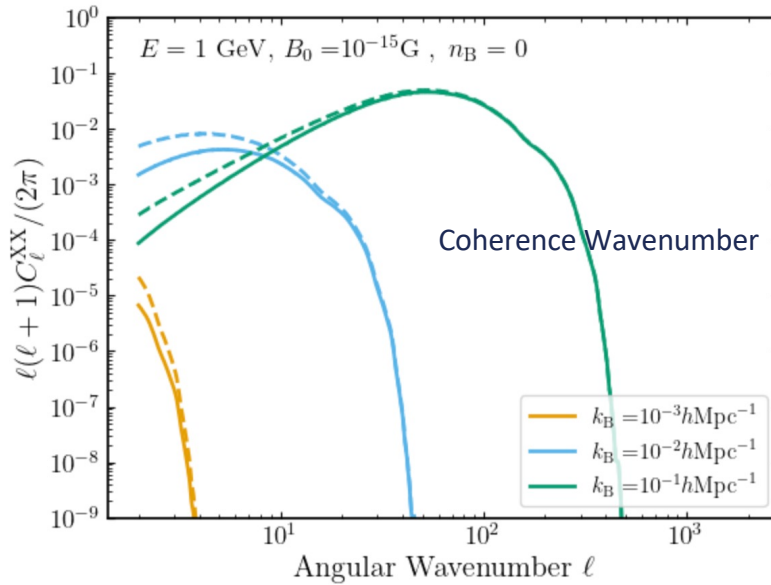
[Schmidt & Jeong (2013)]

E- & B-Mode Angular Power Spectra

$$\mathbf{B}(\mathbf{x}) = \sum_{s=\pm} \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{-i\mathbf{k}\cdot\mathbf{x}} B_s(\mathbf{k}) \hat{\mathbf{e}}_s(\hat{\mathbf{k}})$$

$$\langle B_s(\mathbf{k}) B_{s'}^*(\mathbf{k}') \rangle = (2\pi)^3 \delta_{ss'} \delta_D^{(3)}(\mathbf{k} - \mathbf{k}') P_s(k)$$

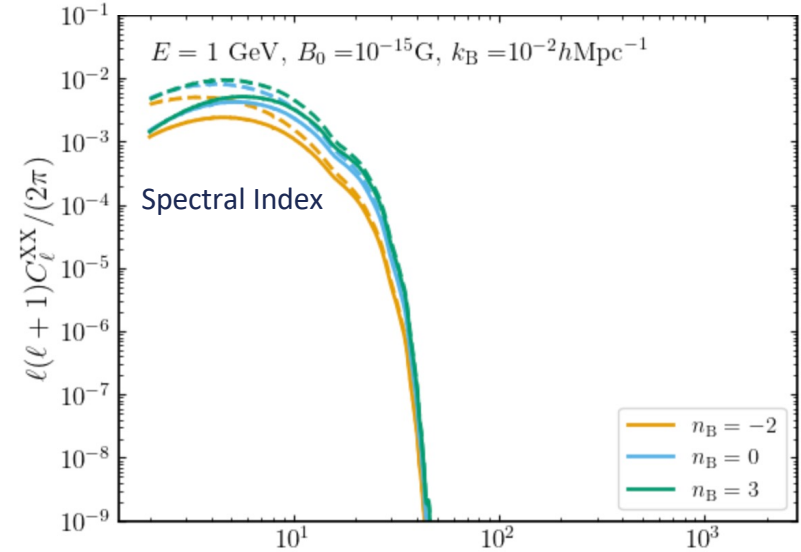
$$P_s(k) = B_s^2 (n_B + 3) \frac{2\pi^2}{k_B^3} \left(\frac{k}{k_B} \right)^{n_B} \Theta(k_B - k)$$



$$C_\ell^{XY} = \frac{I_{XY}}{4} (\ell+2)(\ell-1) \times \left\{ \frac{2}{\pi} \int dk k^2 \Delta_\ell^X(k, E) \Delta_\ell^Y(k, E) \frac{[P_+(k) + \sigma_{XY} P_-(k)]}{2} \right\}$$

$$\Delta_\ell^E(k, E) = \int_0^\infty d\chi \frac{P(\chi, E)}{B_c(\chi, E)} j_\ell(k\chi)$$

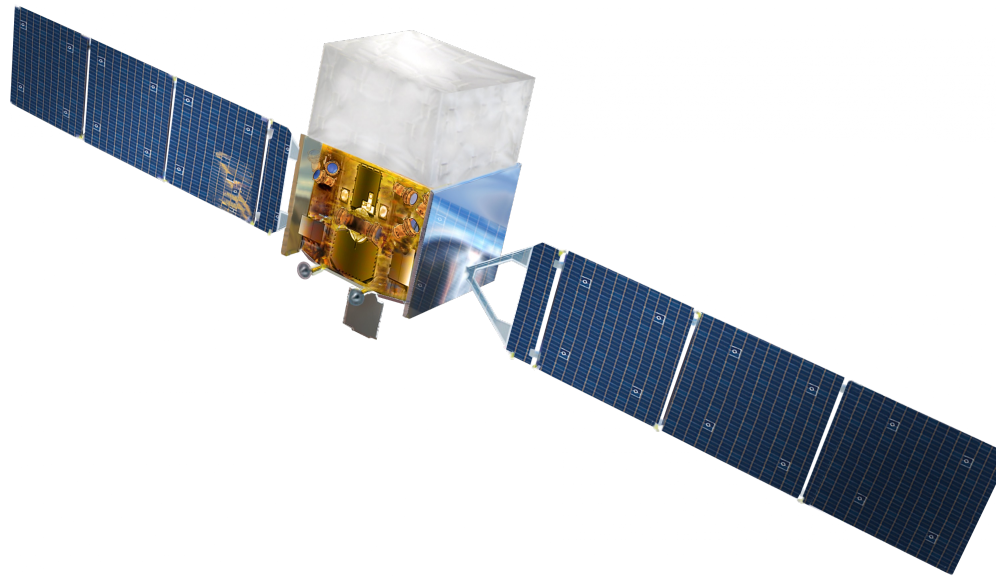
$$\Delta_\ell^B(k, E) = \int_0^\infty d\chi \frac{P(\chi, E)}{B_c(\chi, E)} \left(1 + \frac{2}{k\chi} \right) j_\ell(k\chi)$$



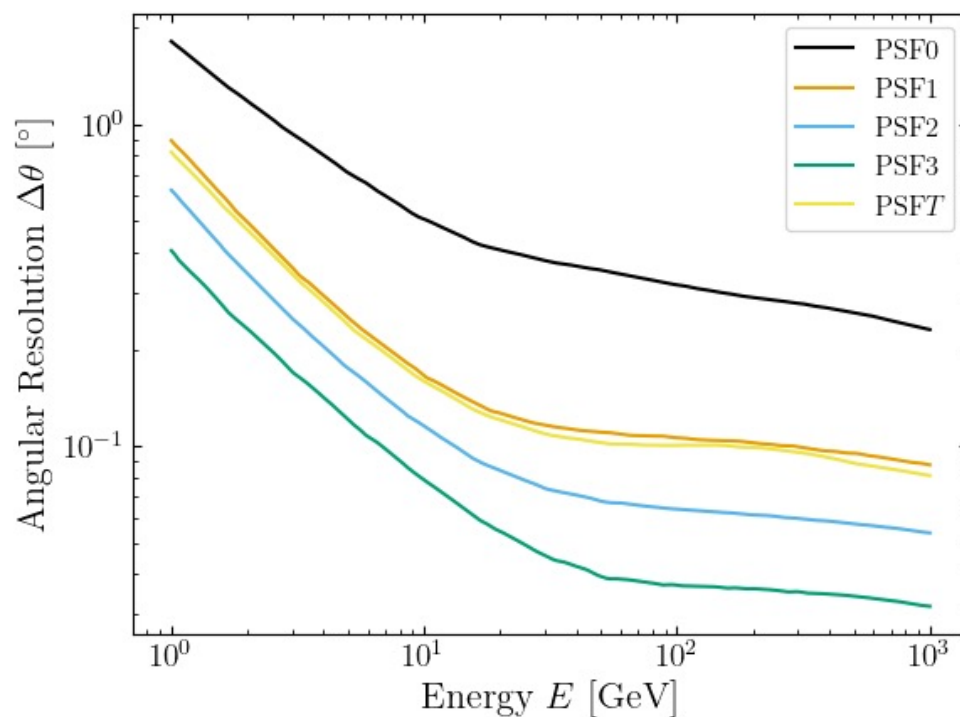
$$\ell_{\text{peak}} = k_B \chi_c \chi_c = 4.45 \text{ Gpc}$$

III. Noise

Fermi Large Area Telescope (LAT)

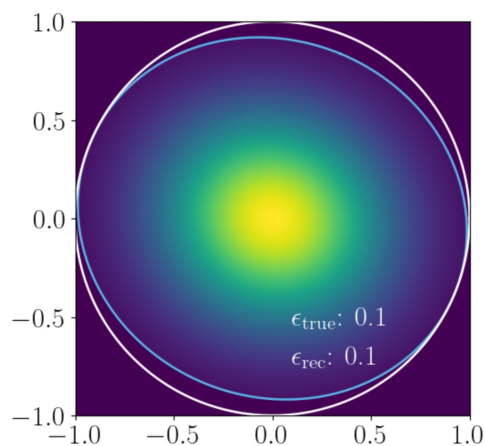


Sources of Error (1/5): Centroid Precision



[https://www.slac.stanford.edu/exp/glast/groups/canda/lat_Performance.html]

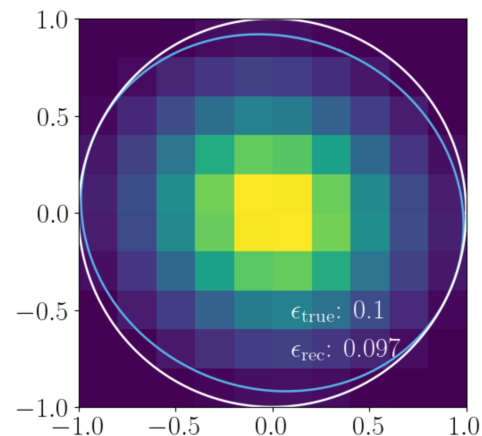
Sources of Error (2/5): Pixelization



Convolution:

$$\gamma_1 = \frac{\epsilon \cos(2\theta)}{2} \frac{1}{1 + (1 + \epsilon)/(12n^2)}$$

$$\gamma_2 = \frac{\epsilon \sin(2\theta)}{2} \frac{1}{1 + (1 + \epsilon)/(12n^2)}$$



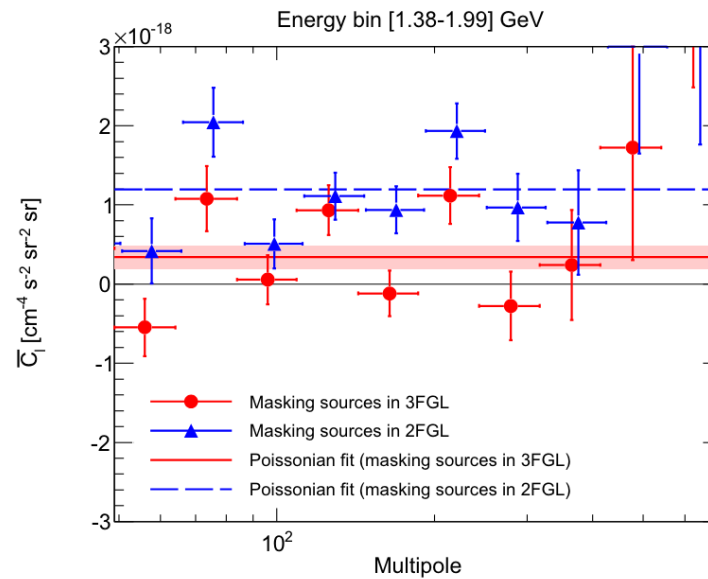
Sampling from Center:

Trapezoidal Rule

$$\mathbf{E} = -\frac{(b-a)^2}{12N^2} [f'(b) - f'(a)]$$

Sources of Error (3/5): Large Foregrounds

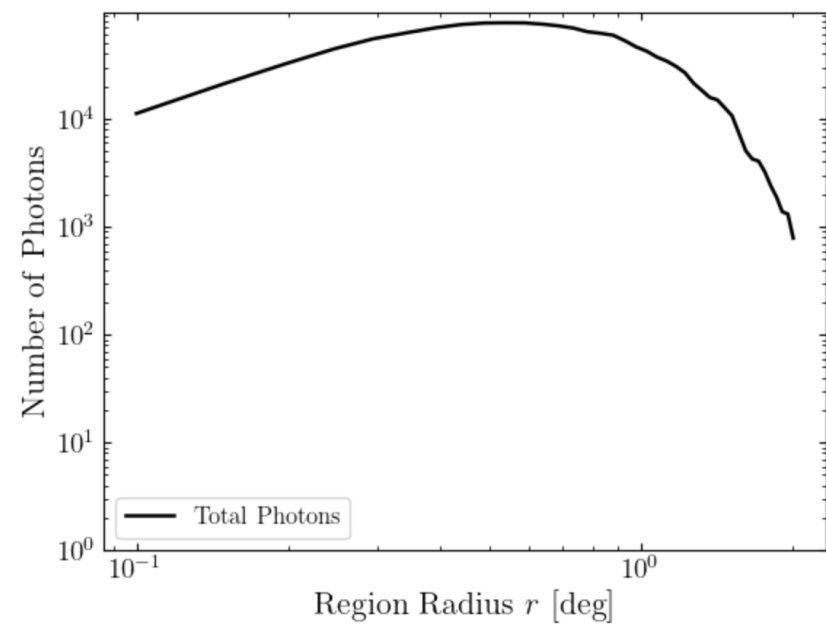
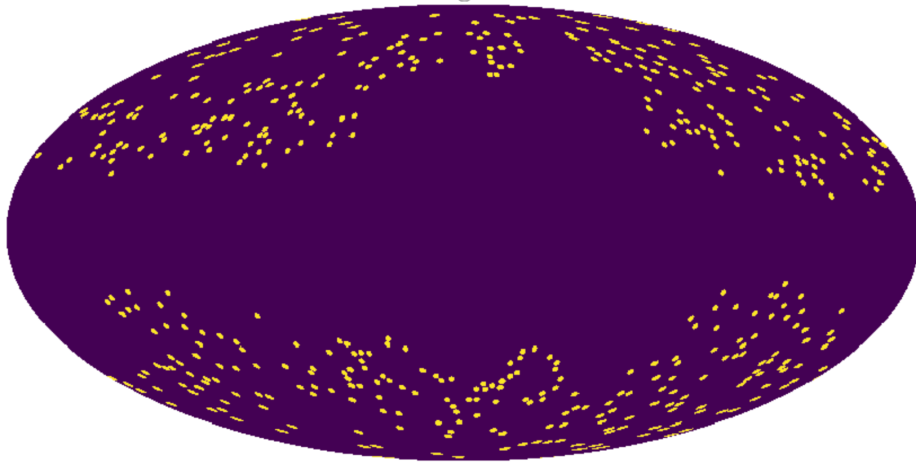
The angular power spectrum of the diffuse gamma-ray emission as measured by the *Fermi* Large Area Telescope and constraints on its Dark Matter interpretation



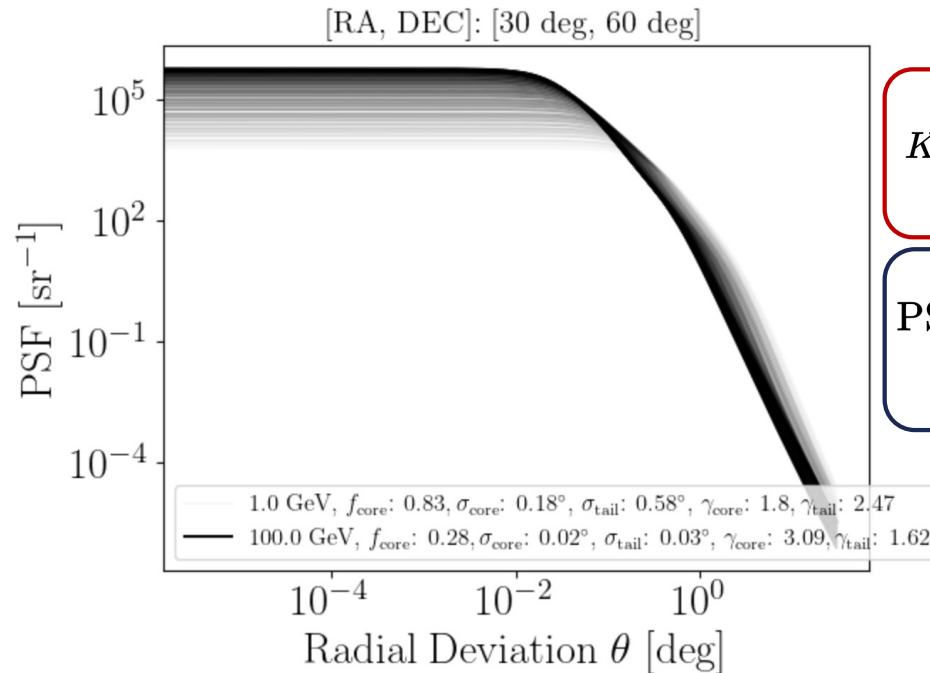
$$\gamma_{\text{obs}} \sim \left[\bar{I}_{\text{src}} / \left(\bar{I}_{\text{src}} + \bar{I}_{\text{bkg}} \right) \right] \gamma_{\text{src}}$$

Sources of Error (4/5): Shot Noise

Lone Blazar Map, $r_{\text{region}} = 1.0^\circ$, $N_{\text{bl}} = 463$



Sources of Error (5/5): PSF Shape



$$K(x, \sigma, \gamma) = \frac{1}{2\pi\sigma^2} \left(1 - \frac{1}{\gamma}\right) \left[1 + \frac{1}{2\gamma} \frac{x^2}{\sigma^2}\right]^{-\gamma}$$

$$\text{PSF} \left(x; \vec{\theta}_{\text{PSF}}\right) = f_{\text{core}} K(x, \sigma_{\text{core}}, \gamma_{\text{core}}) + (1 - f_{\text{core}}) K(x, \sigma_{\text{tail}}, \gamma_{\text{tail}})$$

[Moffat (1969)]

[Read, Rosen, Saxton, & Ramirez (2011)]

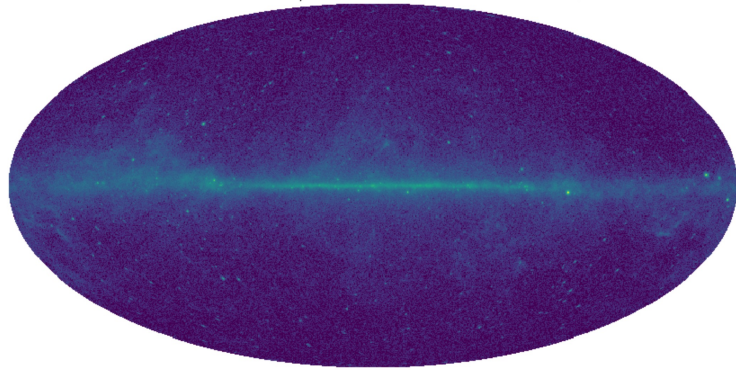
Caveats About Analyzing LAT Pass 8 (P8R3_V3) Data

- The effective area tables are averaged over the azimuthal angle in the LAT instrument reference frame. For time scales of more than a few days this averaging is accurate to better than 1-2%. However, for shorter timescales it is important to use the [optional phi-dependent extensions to the livetime](#) when calculating exposures.

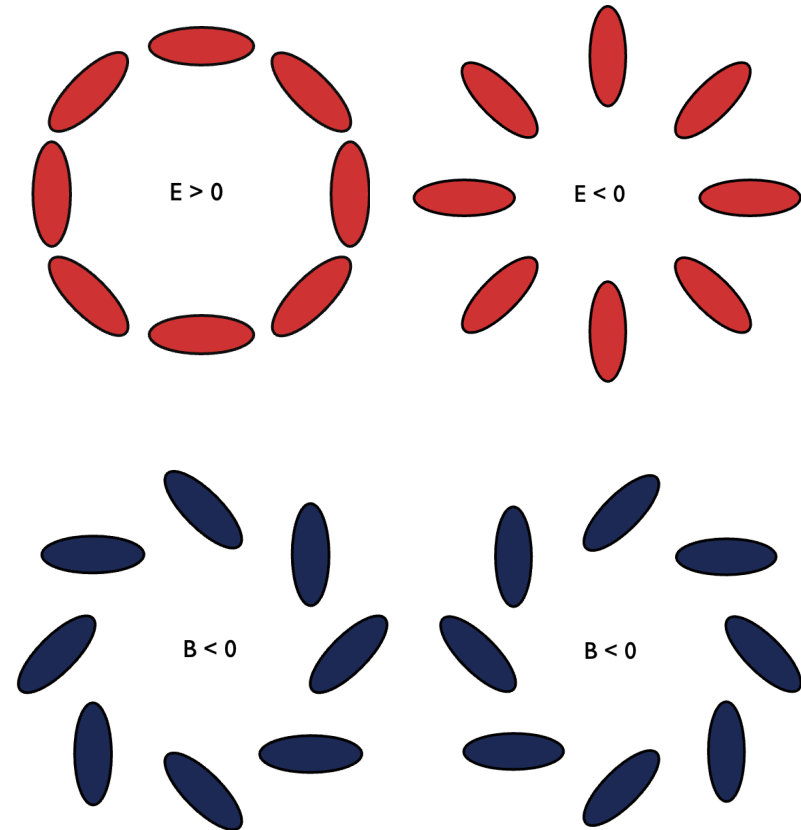
[https://fermi.gsfc.nasa.gov/ssc/data/analysis/LAT_caveats.html]

Null Tests

14 Year Fermi Data w/ GTIs, zenith $< 90^\circ$, 1-100 GeV



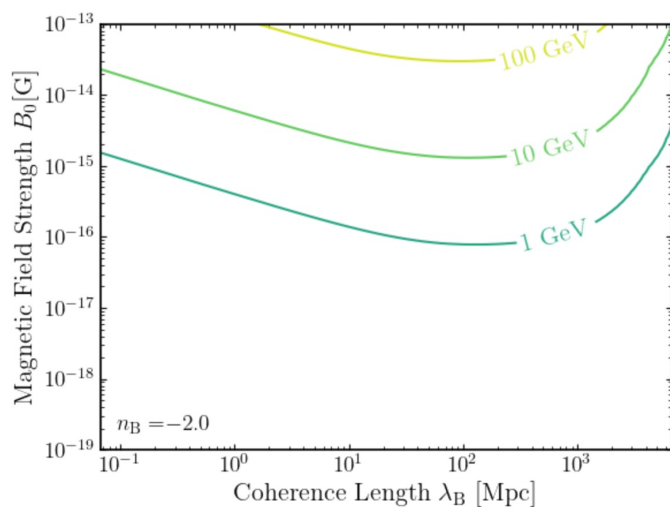
$$R_{ij}[\theta(\hat{n})]\gamma_i(\hat{n})$$



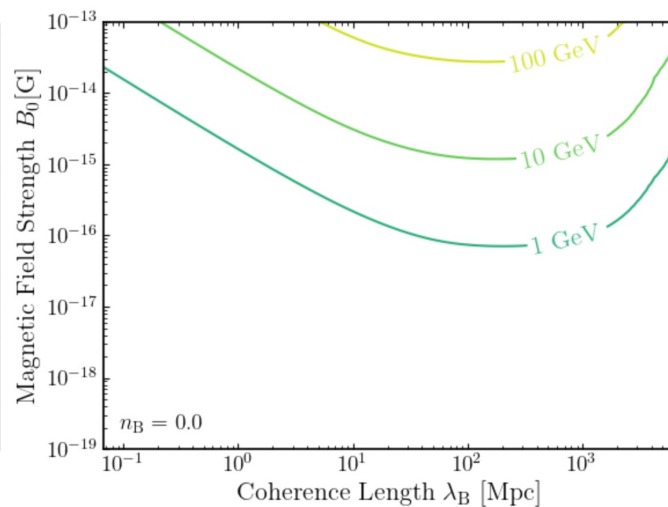
Fermi-LAT Sensitivity

$$\langle \mathbf{B}(\mathbf{x}) \cdot \mathbf{B}(\mathbf{x}) \rangle = B_0^2 \equiv B_+^2 + B_-^2$$

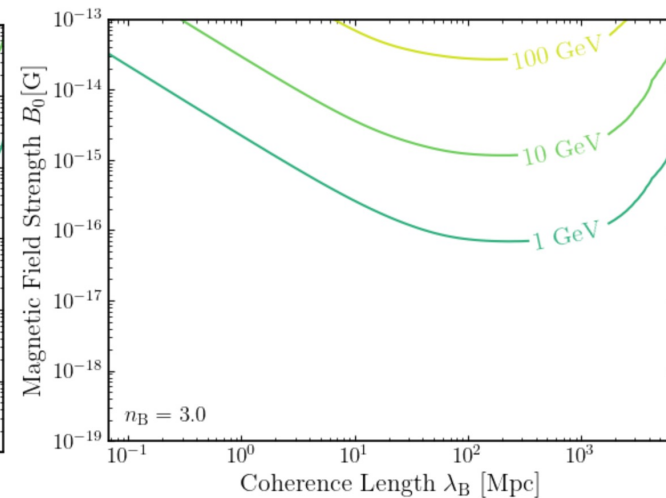
$$\lambda_B \sim 2\pi/k_B$$



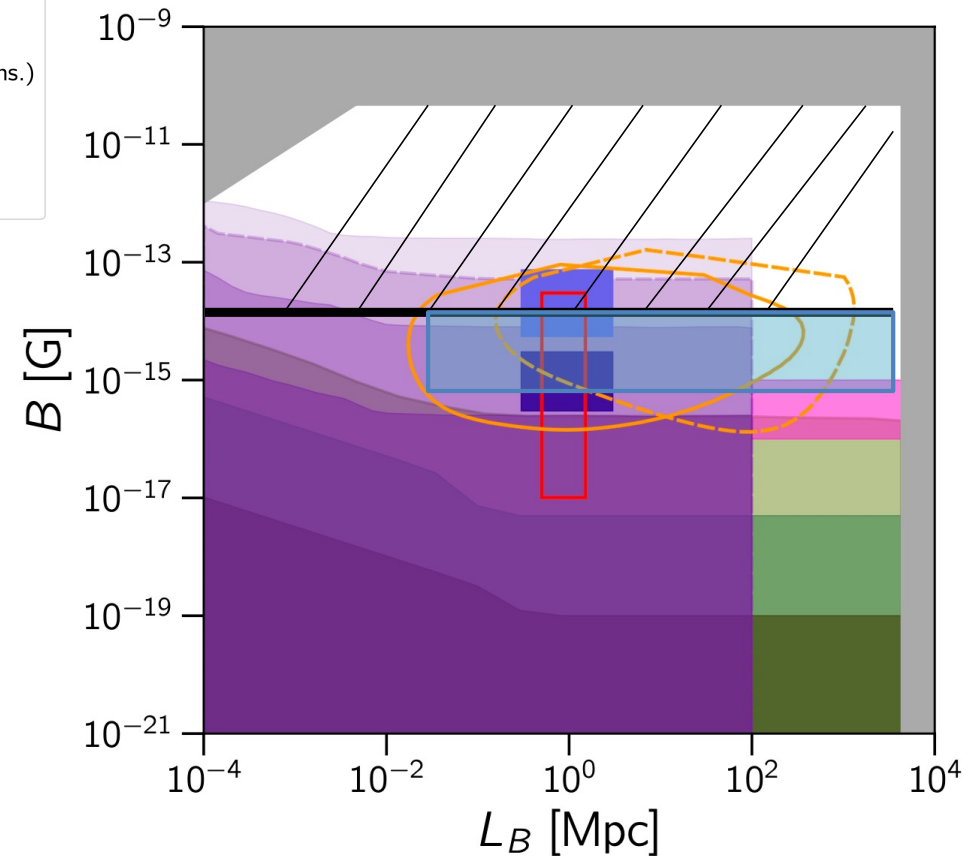
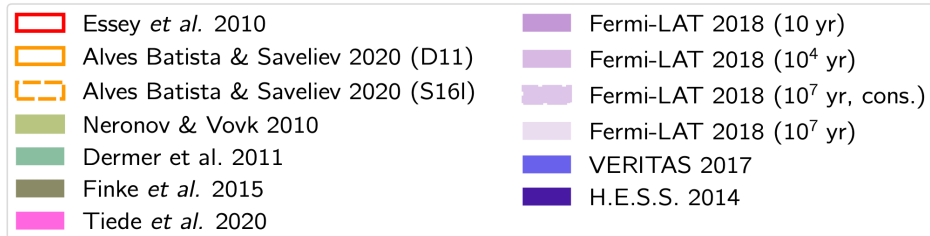
Negative



Zero



Positive



The Gamma Ray Window to Intergalactic Magnetism, Batista & Saveliev: 2105.12020

Where are the Cascades from Blazar Jets? An Emerging Tension in the γ -ray sky, Blanco et al: 2303.01524

IV. Comment

Other Blazar-Based Techniques

1) Stacking Analyses

[Ando & Kusenko (2010)]

[Nerenov (2011)]

[Fermi LAT Collaboration (2013)]

[Fermi LAT Collaboration (2018)]

3) Fermi Handedness

[Kahniashvili & Vachaspati, (2006)]

[Tashiro & Vachaspati (2013)]

[Long & Vachaspati (2015)]

[Duplessis & Vachaspati (2017)]

2) Search for Bow Ties

[Broderick, Tiede, Shalaby, Pfrommer, Puchwein, Chang, & Lamberts (2016)]

[Tiede, Broderick, Shalaby, Pfrommer, Puchwein, Chang, & Lamberts (2017)]

[Broderick, Tiede, Chang, Lamberts, Pfrommer, Puchewin, Shalaby, Werhahn (2018)]

Conclusions

- The presence of an IGMF induces correlated quadrupolar distortions on blazar images (or more generally, TeV sources with distant pair-production).
- Through an E & B-mode decomposition, the associated angular power spectra can distinguish between parity-even and -odd IGMFs.
- Fermi is sensitive to IGMFs with strengths $\sim(10^{-16}-10^{-14})$ G and coherence lengths $\sim(1-100)$ Mpc for a variety of spectral indices.
- The theory modeling of correlated quadrupolar distortions is more accurate and the measurement is more robust to uncertainties compared to other blazar-based methods of IGMF detection.
- More analysis underway!

Thank you.



IGMFs Broadly: What and Why?

- Galactic Magnetic Field Seeds

$$B_{\text{seed}} \gtrsim 10^{-20} \text{G} \rightarrow B_{\text{galaxy}} \sim \mu\text{G}, \quad B_{\text{seed}} = B_{\text{IGMF}}$$

- Baryogenesis Signatures

$$\mathcal{L} \supset \frac{\varphi}{4f} F_{\mu\nu} \tilde{F}^{\mu\nu} \quad \partial_\mu j_B^\mu = \partial_\mu j_L^\mu = N_g \frac{g^2}{16\pi^2} \text{Tr}[W_{\mu\nu} \widetilde{W}^{\mu\nu}] - N_g \frac{g'^2}{32\pi^2} Y_{\mu\nu} \tilde{Y}^{\mu\nu}$$

- Inflationary/Phase Transition Signatures

$$RA^2, R_{\mu\nu} A^\mu A^\nu, RF^{\mu\nu} F_{\mu\nu} \exp(\phi) F_{\mu\nu} F^{\mu\nu}$$

- Costs Nothing

$$A_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \\ -i \frac{2 \sin \theta_w}{g\eta^2} (\partial_\mu \Phi^\dagger \partial_\nu \Phi - \partial_\nu \Phi^\dagger \partial_\mu \Phi)$$

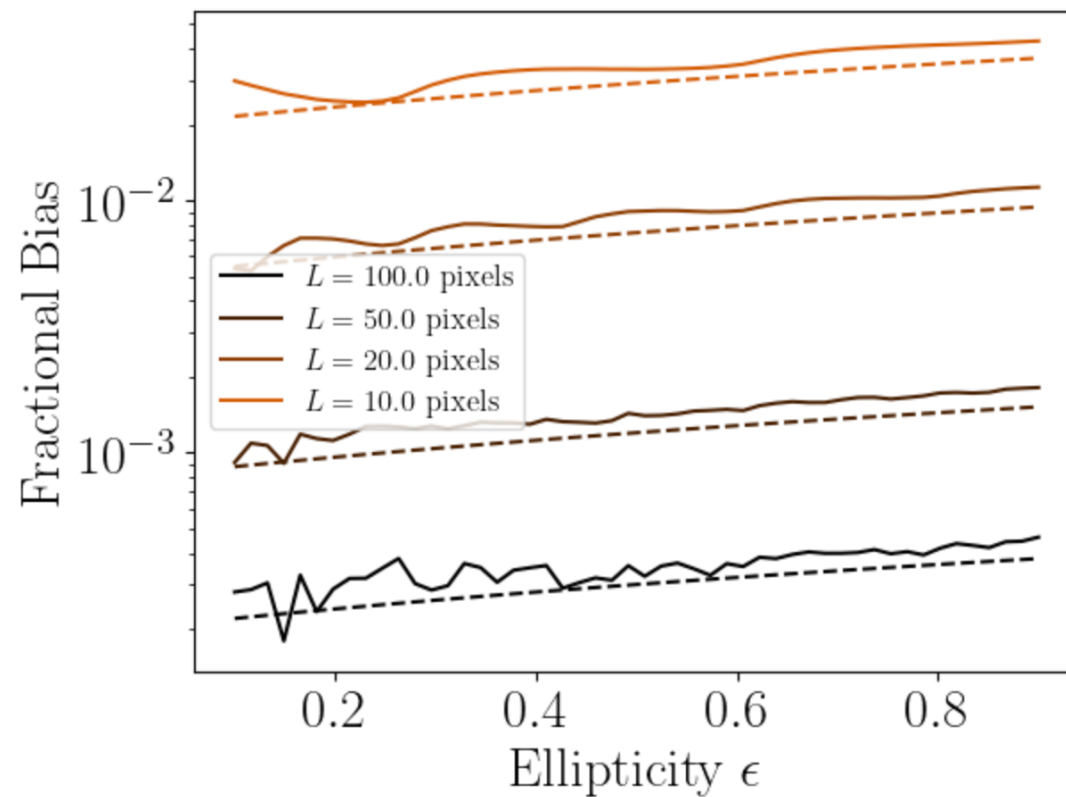
Extra Slide

$$\nabla^2 \varphi(\hat{\mathbf{n}}, E) = \int d\chi \frac{P(\chi, E)}{B_c(\chi, E)} \hat{\mathbf{n}} \cdot [\chi \nabla \times \mathbf{B}(\chi, \hat{\mathbf{n}})] \quad (26)$$

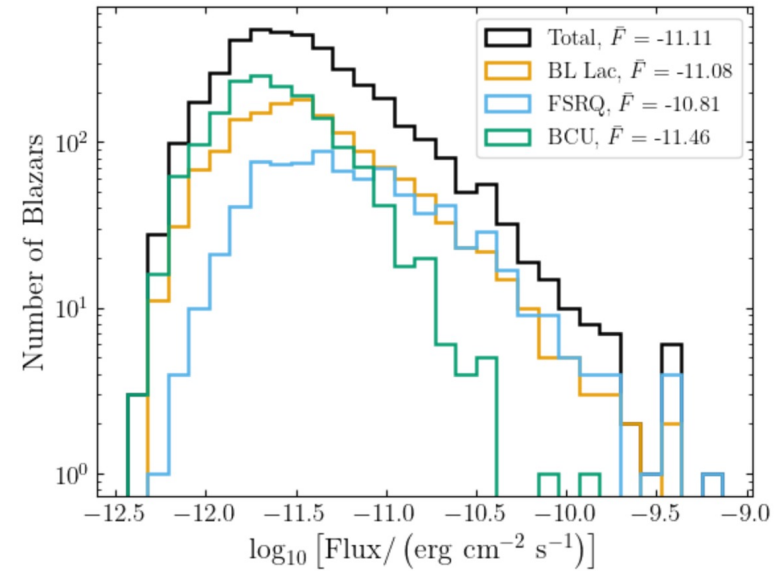
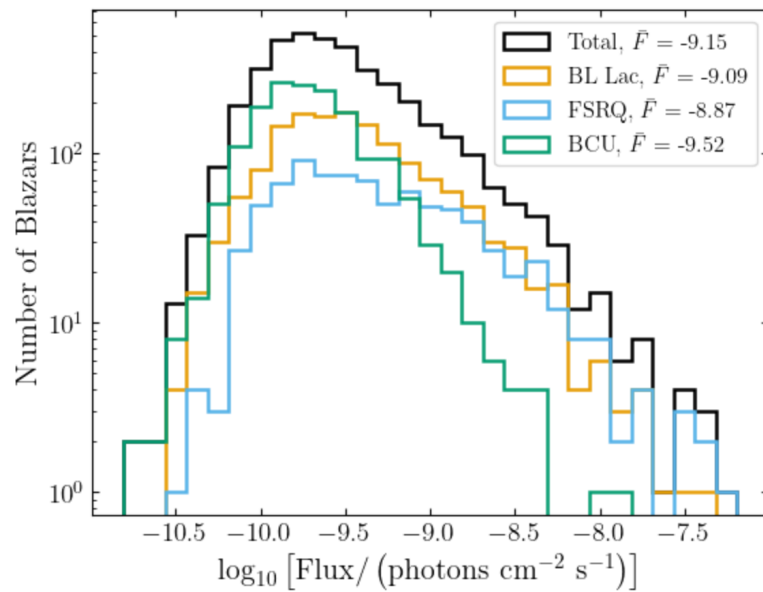
$$\nabla^2 \psi(\hat{\mathbf{n}}, E) = \int d\chi \frac{P(\chi, E)}{B_c(\chi, E)} \hat{\mathbf{n}} \cdot \left(2 + \chi \frac{\partial}{\partial \chi} \right) \mathbf{B}(\chi, \hat{\mathbf{n}}), \quad (27)$$

Pixelization Bias Verification

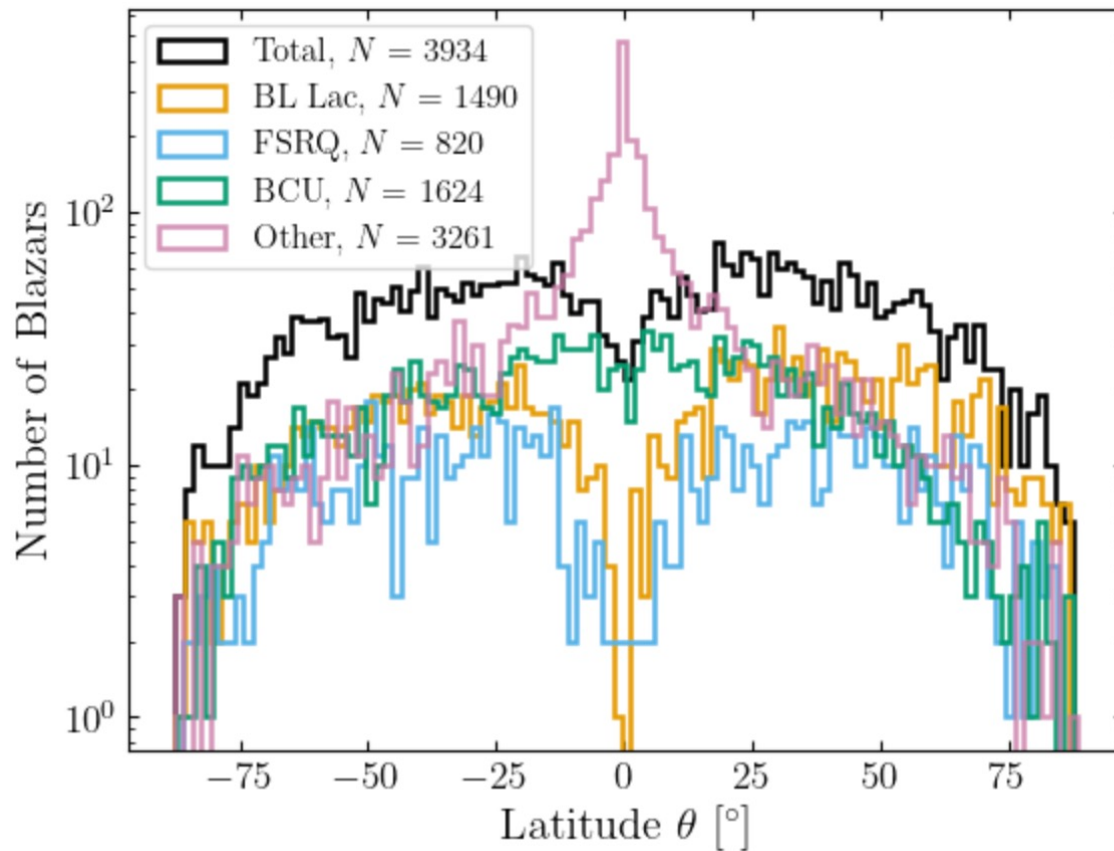
$$\gamma_1 = \frac{\epsilon \cos(2\theta)}{2} \frac{1}{1 + (1 + \epsilon)/(12n^2)}$$
$$\gamma_2 = \frac{\epsilon \sin(2\theta)}{2} \frac{1}{1 + (1 + \epsilon)/(12n^2)}$$



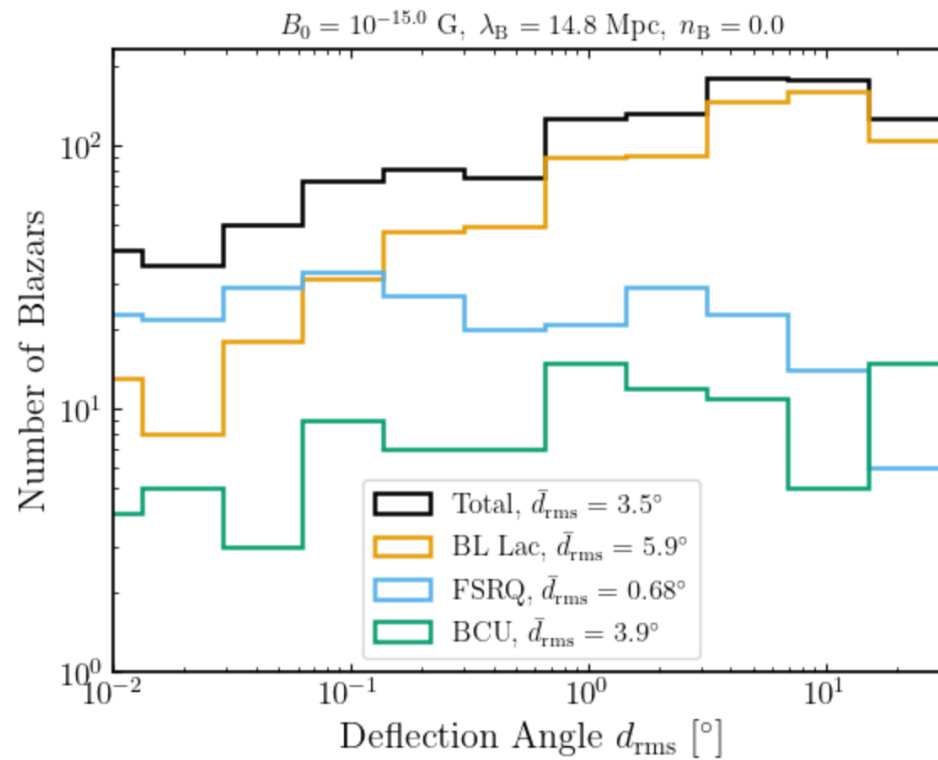
Extra Slide: Blazar Flux Distributions



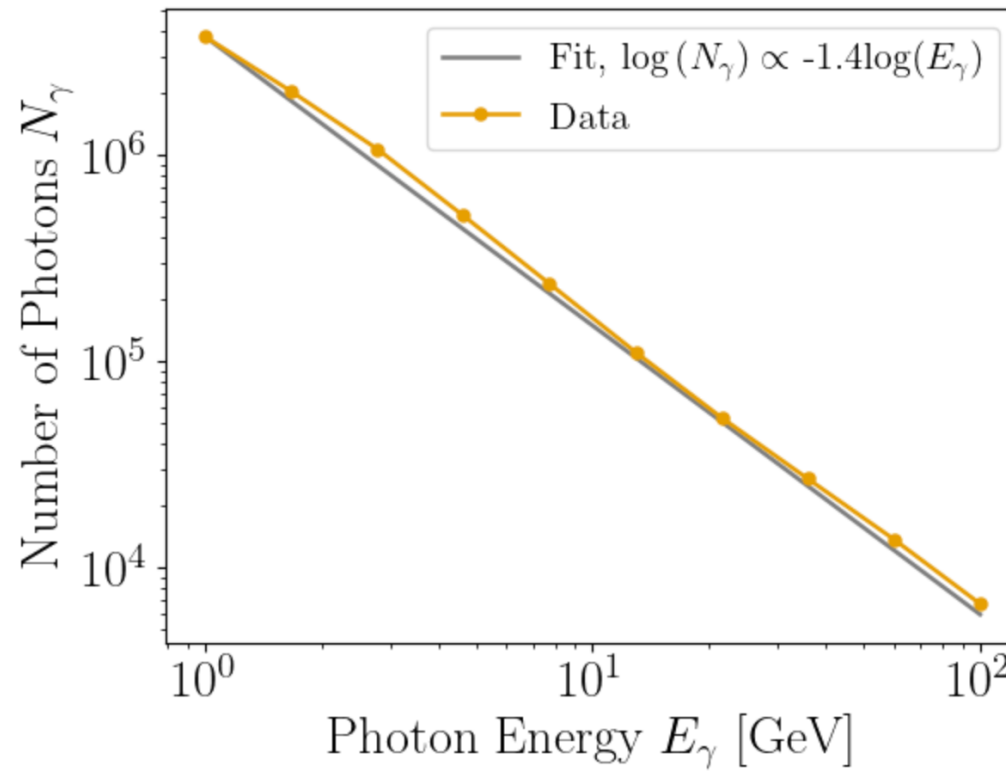
Extra Slide: Blazar Latitude Distributions



Extra Slide: Deflection Angles for Fermi

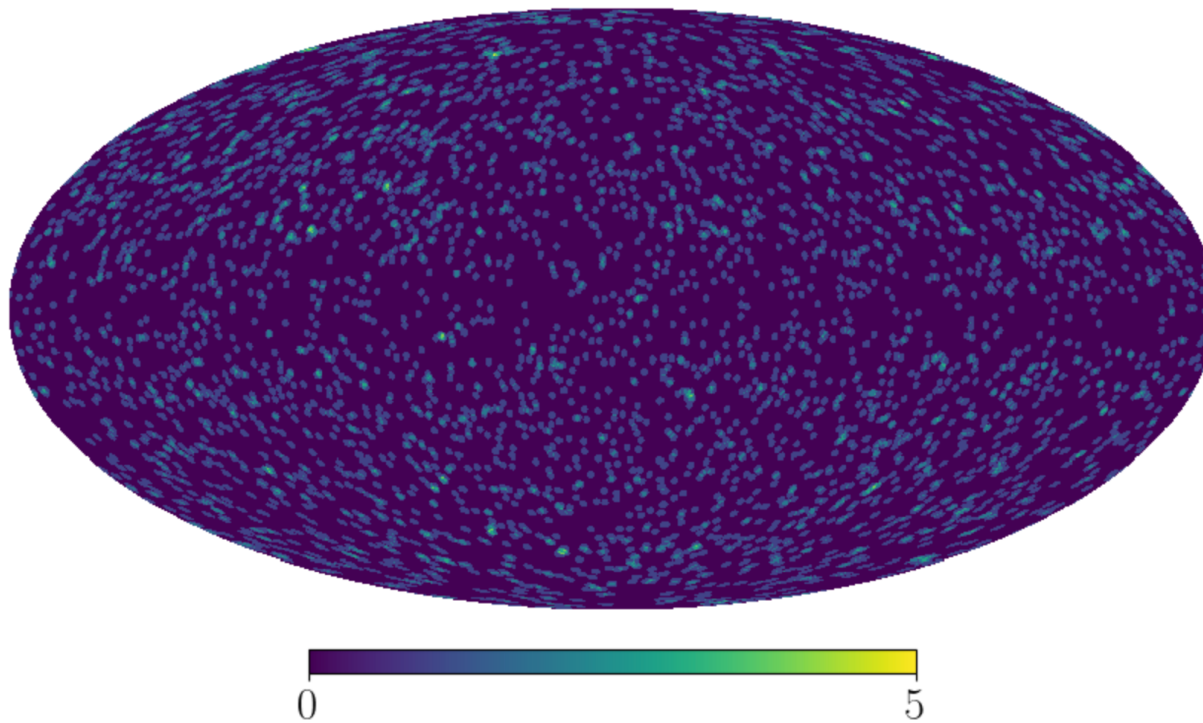


Extra Slide: Total Photon Counts vs Energy



Extra Slide: Pre-Cleaning Blazar Map

Mollweide view



Extra Slide: Resolution of Map

